Environmental Science

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Environmental Science (A01763)

Short Title:	Environmental Science
Faculty	Science and Computing
Department	Science
Cluster	Environmental Science
Author	TWOODCOCK
Credits	5
Level:	Introductory

Description of Module / Aims

This module will give students a broad introduction to the major elements of environmental science, from habitats to the hydrological cycle, to global air pollution issues. A series of laboratory-based experiments plus field trips will be used to reinforce the theory elements of this module.

Programmes

`ENVI-0015`	BSc (Hons) in Agricultural Science (`WD_SAGRI_B`)	2 / 3 / M
`ENVI-0015`	BSc in Agriculture (`WD_SAGUL_D`)	2 / 3 / M

Indicative Content

- Ecology: biodiversity, ecosystems, habitats, niche; species interactions, competition, predation, parasitism, mutualism; succession, climax community; energy flow, food chains & webs
- Water: hydrological cycle; origins of pollution; properties of pollutants; effects of pollution; pollution control, including wastewater and water treatment; EU Water Frameworks Directive
- Air: natural and polluted atmospheres; meteorological factors affecting air pollution; global air pollution issues, i.e. acid deposition, global warming, ozone depletion, air pollution and health effects
- Nutrient cycles, specifically nitrogen and carbon
- Field trips: field trips to river & woodland ecosystems including measurement of biotic index of water quality
- Laboratory practicals: physico-chemical analysis of water, i.e. hardness, BOD, nitrate and phosphate in water; microbiological analysis of water

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the various parameters important in ecosystems.
2. Explain the main topics associated with water in our environment such as the hydrological cycle, pollution & treatment.
3. Outline significant air pollution components and explain their significance in terms of acidity, global warming and ozone depletion.
4. Summarise the importance of nitrogen and carbon cycling.
5. Identify the major ecological and physico chemical parameters necessary to access environmental quality.
6. Classify environments based on laboratory and field investigations.
7. Discuss how pollutants are produced and monitored in the environment.

Learning and Teaching Methods

- Lectures.

- Laboratory investigations plus field trips will reinforce and support theoretical elements of the module.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4
Continuous Assessment	50%	
Lab Report	40%	5,6,7
In-Class Assessment	10%	5,6,7

Assessment Criteria

<40%: No understanding of the basic principles of environmental science.
40%-49%: Is able to understand the principles of environmental science. May have some difficulty with applying this theory.
50%-59%: All the above in addition to evaluation of practical data to assess environmental quality.
60%-69%: In addition to the above plus show a high level of competence and efficiency in the subject area.
70%-100%: All previous to excellent level. Be able to describe and discuss environmental science.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Lab	24	
Independent Learning	87	

Essential Material

- "The EU Environment web site." http://ec.europa.eu/environment. http://ec.europa.eu/environment/index_en.htm
- "The Irish EPA." www.epa.ie. http://www.epa.ie/\#&panel1-1
- Baird, C. and M. Cann. _Environmental Chemistry_. 5th Edition. New York: W.H. Freeman Company, 2012.
- Jones, A.M. _Environmental Biology_. UK: Routledge, 1997.

Supplementary Material

- "American EPA." www.epa.gov. http://www.epa.gov/
- "Intergovernmental Panel on Climate Change." www.ipcc.ch. http://www.ipcc.ch/
- Chiras, D. _Environmental Science_. 9th Edition. US: Jones and Bartlett, 2012.
- Falkner, R. _The handbook of global climate and environment policy_. Chichester, U.K.: Wiley-Blackwell, 2013.

- Lehane, M. and B. O'Leary. _Ireland's Environment: An Assessment_. Ireland: EPA Publications, 2012.
- McKinney, M.L., R.M. Schoch and L. Yonavjak. _Environmental Science_. 5th Edition. US: Jones and Bartlett, 2013.